nature plants

Letter

https://doi.org/10.1038/s41477-023-01541-7

Widespread synchronous decline of Mediterranean-type forest driven by accelerated aridity

Received: 11 July 2022

Accepted: 12 September 2023

Published online: 16 October 2023

Check for updates

Alejandro Miranda[1,2] ✉, Alexandra D. Syphard[3,4], Miguel Berdugo[5], Jaime Carrasco[6], Susana Gómez-González[2,7,8], Juan F. Ovalle[9,10], Cristian A. Delpiano[11,12], Solange Vargas[13], Francisco A. Squeo[11,12], Marcelo D. Miranda[10,14], Cynnamon Dobbs[15], Rayen Mentler[2], Antonio Lara[2,16,17] & René Garreaud[2,18]

Large-scale, abrupt ecosystem change in direct response to climate extremes is a critical but poorly documented phenomenon[1]. Yet, recent increases in climate-induced tree mortality raise concern that some forest ecosystems are on the brink of collapse across wide environmental gradients[2,3]. Here we assessed climatic and productivity trends across the world's five Mediterranean forest ecosystems from 2000 to 2021 and detected a large-scale, abrupt forest browning and productivity decline in Chile (>90% of the forest in <100 days), responding to a sustained, acute drought. The extreme dry and warm conditions in Chile, unprecedented in the recent history of all Mediterranean-type ecosystems, are akin to those projected to arise in the second half of the century[4]. Long-term recovery of this forest is uncertain given an ongoing decline in regional water balance. This dramatic plummet of forest productivity may be a spyglass to the future for other Mediterranean ecosystems.

Ecosystem collapse involves an abrupt change in conditions that fundamentally alters one or more of an ecosystem's defining characteristics beyond the threshold of natural variability[5–7]. In terrestrial ecosystems, direct climate-driven collapse, such as vegetation type conversion or extinction of native biota, has thus far manifested mostly at limited geographical extents[8]. While less commonly documented, macro-scale ecosystem degradation and collapse may occur as a result of long-lasting, decadal, recurrent and/or intense droughts[1]. Since droughts of such extremity have rarely occurred, their capacity to drive notable biodiversity shifts remains largely unknown[1].

[1]Laboratorio de Ecología del Paisaje y Conservación, Departamento de Ciencias Forestales, Universidad de La Frontera, Temuco, Chile. [2]Center for Climate and Resilience Research (CR2), Santiago, Chile. [3]Department of Geography, San Diego State University, San Diego, CA, USA. [4]Conservation Biology Institute, Corvallis, OR, USA. [5]Institute of Integrative Biology, Department of Environment Systems Science, ETH Zurich, Zürich, Switzerland. [6]Departamento de Industria, Facultad de Ingeniería, Universidad Tecnológica Metropolitana, Santiago, Chile. [7]Departamento de Biología-IVAGRO, Universidad de Cádiz, Puerto Real, Spain. [8]Center for Fire and Socioecological Systems (FireSES), Universidad Austral de Chile, Valdivia, Chile. [9]Facultad de Ciencias Forestales y de La Conservación de la Naturaleza, Universidad de Chile, Santiago, Chile. [10]Center of Applied Ecology and Sustainability (CAPES), Santiago, Chile. [11]Departamento de Biología, Facultad de Ciencias, Universidad de La Serena, La Serena, Chile. [12]Instituto de Ecología y Biodiversidad (IEB), Santiago, Chile. [13]Departamento de Química y Biología, Facultad de Ciencias Naturales, Universidad de Atacama, Copiapó, Chile. [14]Facultad de Agronomía e Ingeniería Forestal, Pontificia Universidad Católica de Chile, Santiago, Chile. [15]Department of Natural Resources and the Environment, University of Connecticut, Mansfield, CT, USA. [16]Instituto de Conservación, Biodiversidad y Territorio, Universidad Austral de Chile, Valdivia, Chile. [17]Fundación Centro de los Bosques Nativos FORECOS, Valdivia, Chile. [18]Departamento de Geofísica, Universidad de Chile, Santiago, Chile. ✉e-mail: alejandro.miranda@ufrontera.cl

Therefore, assessments of climate-driven ecosystem change mostly rely on space-for-time substitution approaches, results from which are difficult to validate or reproduce over time[9].

The shrublands and forests of the world's five Mediterranean-type ecosystems (MTEs; Supplementary Fig. 1) are especially important to monitor for signs of ecosystem change because they are areas of high species richness and biodiversity that may be particularly vulnerable to hot, drought-driven ecosystem response given their adjacency to semi-arid and desert ecosystems. Although woody species of MTEs are generally adapted to low soil water potential[10], changes in climatic extremes may exceed species physiological tolerances[11–13]. Large-scale ecosystem change in MTEs could have serious socio-ecological impacts because these biodiversity hotspots also host ~500 million people[10]. Understanding and anticipating the direct role of climate change on regional ecosystem change is crucial to develop adaptation and mitigation programmes that may prevent or reduce their impacts. Here we analysed trends in forest productivity using remote sensing vegetation indices, in addition to a quantification of the climatic context, in the last two decades across the world's five MTEs to investigate signs of large or abrupt ecosystem change. Detecting dramatic changes could help to monitor for signs of vulnerability or potential for subsequent ecosystem collapse.

Satellite-based normalized difference vegetation index (NDVI) time series were used as a surrogate for dominant tree layer photosynthetic productivity or greenness[14,15], to assess 21 years of change in ~118,000 km^2 of MTE forests worldwide (Supplementary Fig. 1 and Table 1). We found no significant negative trends in the regional annual mean NDVI across all MTEs, but there was an abrupt and significant ($P < 0.01$) decline in Central Chile not observed in the other MTEs (Fig. 1a). For the years 2019, 2020 and 2021, we observed a decrease in regional NDVI annual mean values compared with 2000–2010 of 12.1%, 19% and 11.6%, respectively. In 2020, Chile experienced the lowest NDVI values and largest negative anomalies of all MTEs across the past couple of decades. After 2019, every year was below the range of variability relative to the same reference (estimated as regional annual mean NDVI ± 2 s.d.; ±5.54%) (Fig. 1b). We also observed local negative NDVI trends in other MTE localities (individual pixel analysis) ($P < 0.01$) between 2000–2021 in California, Southwest Australia, South Africa and the Mediterranean Basin, affecting 8.3%, 6%, 5.7% and 1% of the forest area, respectively. By comparison, 39% of the forest in Chile experienced a declining trend in productivity, revealing the largest impacted area (2,363 km^2) of the MTEs (Supplementary Table 1). A detailed analysis of Central Chile considering only high canopy coverage areas to avoid confounding spectral signal (1,040 km^2) revealed that in late 2019, more than 90% of the region's forest experienced an abrupt decline (mean and maximum anomaly, respectively −29.6% and −57% of 2018 values). These abrupt changes occurred synchronously across 82.4% of the forest within a 100 d time span. The last maximum values of NDVI occurred between June and August 2019 and the minimum NDVI values occurred between February and March 2020. Moreover, these changes occurred across a latitudinal range of ~500 km and ~120 km from the coastal to Andean range, under different local environmental conditions, species composition, structure, dominance and degradation history.

Except for the Mediterranean Basin, the past decade (2010–2021) was drier than average in all MTEs, with mean deficits ranging from −2% (Southwest Australia) to −28% (Central Chile; Fig. 2a). Dry years were dominant but were interspersed with wet years for all MTEs except for Central Chile, where all years since 2010 were below average (Fig. 2a). This conforms with the so-called megadrought affecting a large portion of central-southern Chile (beyond the MTE sector), resulting in the driest decade in local history since at least the 14th century[16] and for any of the MTEs since 1901 (Fig. 2b). Moreover, 2011–2021 was the warmest registered decade (since recorded history) for all MTEs (Fig. 2b), leading to a negative trend in decadal surface water balance expressed as the Standardized Precipitation–Evapotranspiration Index (SPEI-12; Supplementary Fig. 2). The 2010–2020 SPEI-12 averages were exceptionally low across MTEs worldwide, with the lowest being in the Mediterranean Basin and Central Chile (Supplementary Fig. 2).

The climatic anomalies during the past decade in the five MTEs are broadly consistent with projections from coupled general circulation models[17]. In Central Chile, natural variability originating in the subtropical southwest Pacific was conducive to dry conditions across the Pacific[18], acting consistently with climate-change signal. As a result, the observed mean precipitation deficit (~30% of the long-term historical mean) was about three times larger than expected for the early decades of the twenty-first century, more akin to those predicted for 2070–2090 under a heavy greenhouse gas (GHG) emission scenario (Fig. 2a)[4]. The Central Chile megadrought also encompassed 2019, when the region received less than 30% of the mean rainfall (Fig. 2a), accompanied by record high temperatures in spring/summer[19]. This extremely dry and warm year, termed a 'hyper-drought' within the decade-long megadrought, was probably responsible for the sudden and massive forest decline. Even in the case of the decade-scale megadrought that occurred in Australia during pre-industrial time (1890–1900), a sudden evergreen vegetation browning as extensive as this one in Chile has no comparable record in any of the MTEs[1]. Furthermore, while the impacts observed in the Mediterranean ecosystems of Australia at that time were caused by the combination of drought and overgrazing (by exotic rabbits), the vegetative impacts of this drought in Central Chile extended across a wide range of land uses and degradation conditions. In California and the Mediterranean Basin, there has been documentation of large-scale tree mortality and shrubland dieback under recent persistent droughts[20,21]. In Southwest Australia, a sudden canopy collapse of eucalypt forest has been documented, with 26% stem mortality[22]. A recent study of Mediterranean forests in California has shown decline in NDVI to be a strong predictor of subsequent tree mortality[14], which may be a foreboding sign for Chile's forests. In this study, we observed an extraordinary event of forest decline, given its synchronous, abrupt and extensive nature, that may constitute one of the most notable examples of direct climate-driven response in this era of rapid global change for MTEs.

To further investigate local dependencies of observed forest decline, we complemented satellite observations with field measurements of leaf area index (LAI) during 2017–2020 for 474 trees in 78 plots of Central Chile's four most abundant evergreen forest species (Supplementary Fig. 3). We documented responses of species adapted to mesic (*Cryptocarya alba* and *Peumus boldus)* and xeric conditions (*Quillaja saponaria* and *Lithaea caustica*) (Supplementary Table 2). LAI analysis confirmed a significative decrease in productivity ($P < 0.05$; Supplementary Table 3), with the most intense decline observed in trees adapted to more mesic conditions (Supplementary Fig. 4). The distribution of tree crown decay of species groups was bimodal, indicating peaks of either low-level (0–25%) or high-level (75–100%) percentages of canopy damage (Supplementary Fig. 4). This differential response could indicate a yet-unknown physiological mechanism, micro-site condition or density-dependent process that imparts selective resilience to sustained hot-droughts. This may also suggest a rapid massive decline with little intermediate effect (Fig. 3), which could produce significant changes in community composition. In general, extreme drought increases the risk of hydraulic failure and carbon starvation[12,13], but failure of the plant hydraulic system is probably the main mechanism involved in extensive canopy browning[12]. This drought response mechanism may be particularly relevant for evergreen species that exhibit comparatively lower wood density and higher specific leaf area[20]. It is clear that drought weakens trees by the mechanisms indicated; however, it has also been suggested that pathogen attacks and other factors may be more important in directly killing the plants[23]. Nevertheless, this complementary explanation is not evident in our case. Future studies should address whether the variation in drought resistance and browning among tree species is

https://doi.org/10.1038/s41477-023-01541-7

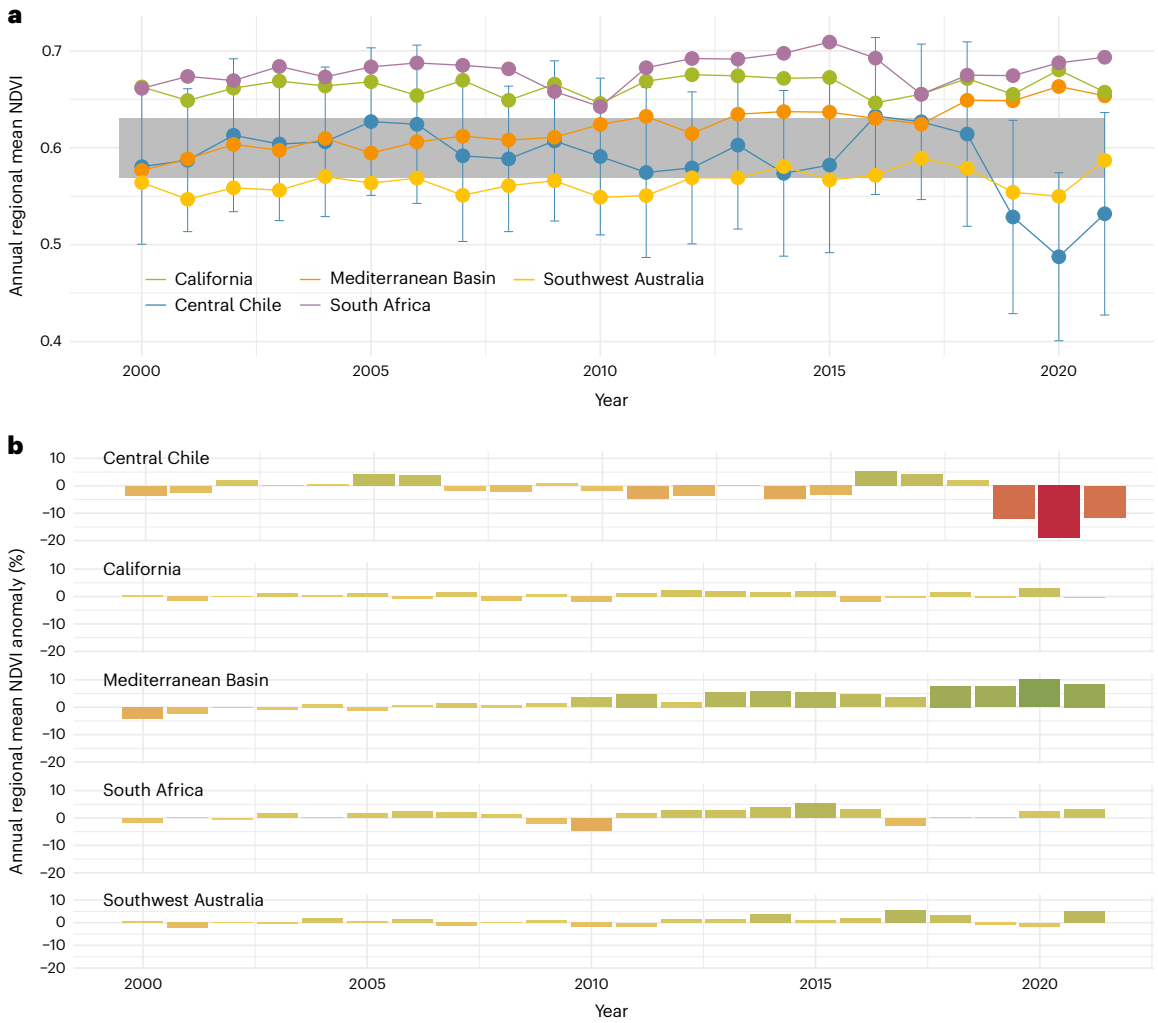


Fig. 1 | Annual regional mean NDVI time series and anomaly for all MTEs. **a**, The absolute annual variation in NDVI for all MTEs represents the mean NDVI value for the entire year across every pixel. The regional mean NDVI was calculated as the mean of the pixels' annual means. The grey band represents the regional annual mean ± 2 s.d. for Central Chile between 2000 and 2010 as a pre-megadrought decadal variability in NDVI values. Error bars for Central Chile indicate the all-seasons variability for every pixel in the region. **b**, Annual regional mean NDVI anomaly for each MTE. Green, positive anomalies; red, negative anomalies. Time series (*N*): California, 16,088; Mediterranean Basin, 422,209; Southwest Australia, 108,679; Central Chile, 33,133; South Africa, 11,525.

ultimately translated into differential survival and a long-term change of community assemblage, causing a long-term regime shift in the community composition.

As in other MTEs, historical land-use change in Central Chile has removed much of the original forests, thereby restricting many of the remaining forests to areas with steeper slopes that have limited access to groundwater, with reduced effect near zones of topographic water accumulation[24]. Moreover, changes in forest structure by land-use history (for example, wildfires, charcoal production and livestock)[24] could reinforce/magnify forest responses to extreme drought episodes. One of the last strategies for drought persistence in evergreen sclerophyllous plants is dying of leaves, branches or the whole canopy[24], which might explain the forests' synchronous decline caused by exceeding the critical threshold of forest water potential. Plant recovery after drought (as after fire) often depends upon reserves stored in the survival organs and the differential ability to sprout from basal buds (for example, lignotubers, roots, rhizomes, or burls). Although regrowth from underground structures is possible given sufficient carbon stocks, resprouting can fail due to hydraulic limitations[25] and, thus, the recovery of these forests is uncertain if drought conditions persist and water balance continues to decrease. Seedling recruitment after disturbance in Chilean sclerophyllous forests is limited by the rapid establishment of exotic plants and high abundance of herbivores[26]. Therefore, if this massive browning turns into mortality by resprouting failure, strong consequences on biodiversity and community structure at regional scale are expected.

In terrestrial ecosystems, direct climate-driven change manifests primarily as local state changes at limited geographical extents, such as vegetation type conversion, local extinction or extensive tree mortality[8]. The relatively fewer examples of abrupt regional-scale shifts mostly result from indirect effects of extreme and punctual climate-driven changes, such as insect or other pathogen outbreaks, altered fire regimes, floods, storms or changes in sea level[2,3,8]. At the local scale, climate-induced forest/tree decline exhibited a patchy spatial pattern determined by plant age, species traits and demographics, disturbance response and composition in combination with, or interacting with, site variability (for example, soil properties and topography)[2,3]. Although the ultimate impacts of the massive 2020 sudden decline in Chile remain to be seen, it has been suggested that large-scale browning could be the initial ecosystem response to increasing aridity and the first step in a cascade of abrupt changes affecting other ecosystem functions and services[9]. Large-scale browning in Chile

https://doi.org/10.1038/s41477-023-01541-7

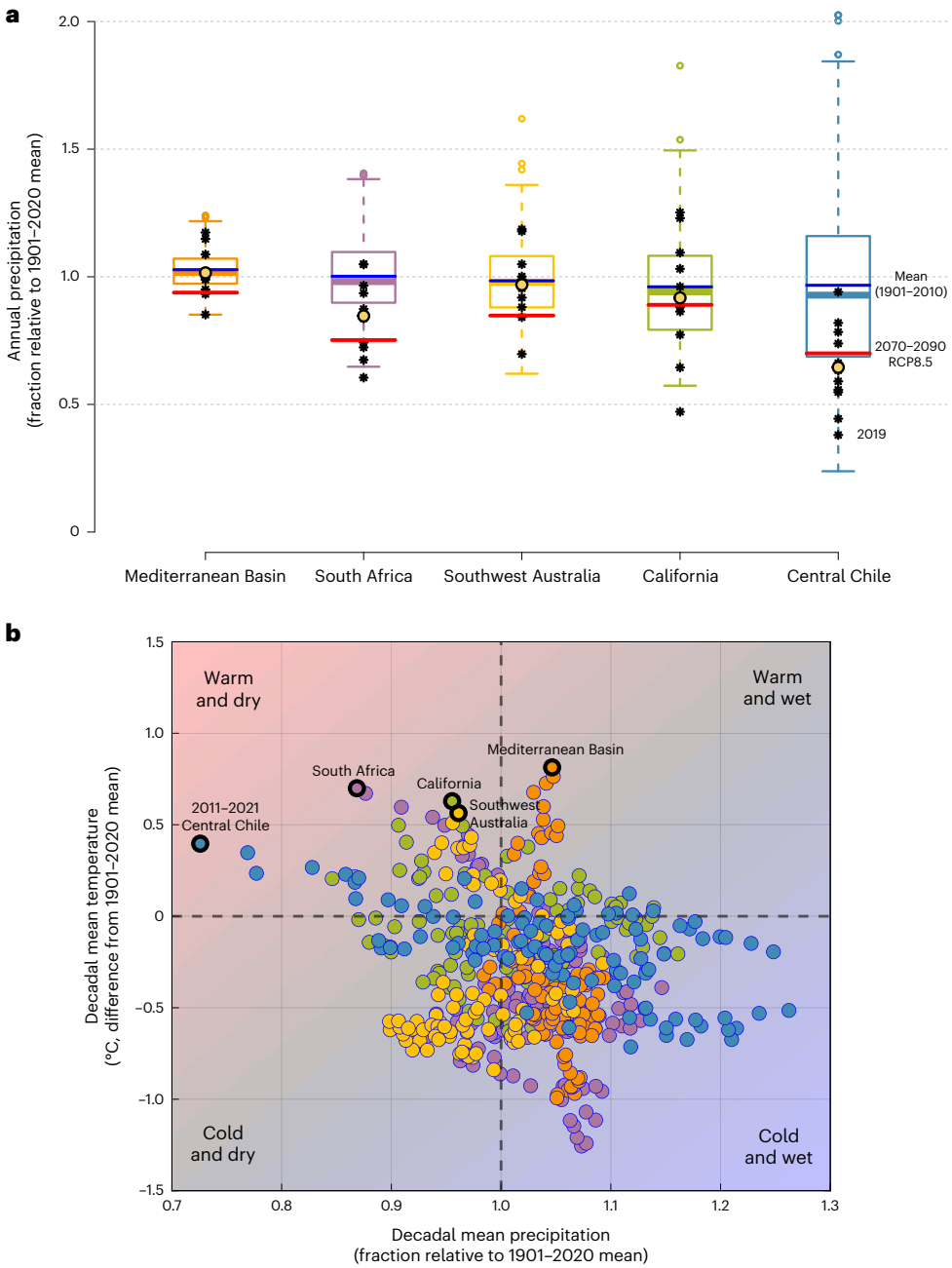


Fig. 2 | MTEs' precipitation and temperature anomalies. a, Boxplot of the annual precipitation in each MTE (blue, Central Chile; purple, South Africa; yellow, Southwest Australia; green, California; orange, Mediterranean Basin) across the period 1901–2020 using CRU data. To compare among MTEs, individual values were divided by the long-term mean (1901–2010) of each region. Values smaller (larger) than one indicate conditions dryer (wetter) than average. The upper (lower) limit of the boxes corresponds to the 75% (25%) quantile; the central line in each box corresponds to the median. Asterisks show the annual precipitation for each year in the period 2010–2020 and the yellow filled circles indicate the 2010–2020 mean. Red lines indicate the expected value of average precipitation for 2070–2090 under RCP8.5. **b**, Plot of decadal mean (1901–1910, 1902–1911, ..., 2011–2020) temperature and precipitation for the five MTEs colour coded as in (**a**). The 2011–2020 means are outlined in black. To compare among regions, each decadal mean precipitation was divided by the long-term mean (1901–2010) of each region (unitless). The temperature anomalies correspond to the decadal mean values minus the long-term climatology (°C). $N = 120$.

may contribute to increasing risk of wildfires[27] which, combined with hot-drought conditions, may arrest natural recovery, potentially pushing the ecosystem to a new stable-state condition[28]. Also, slowing tree growth in the past decades with a potential minimum in 2020 reflects the forest's decreasing and uncertain atmospheric carbon uptake[29]. All these factors operating simultaneously could contribute as positive feedback to the rapid aridification and shifting of this MTE, facilitating disturbances that further promote ecosystem collapse[30].

Model simulations under heavy GHG emission scenarios consistently predict aridification of MTEs, with precipitation reduction of up to 40% and temperature increase of up to 3 °C over the twenty-first century relative to past history[17]. Large-scale, direct climate-driven processes and responses are rare, but ones such as this event in Chile reveal the potential impact of sustained climate change[1,31]. Studies suggest that long-term reduction in productivity and survival of Mediterranean species could reach a tipping point, leading to abrupt, irreversible

https://doi.org/10.1038/s41477-023-01541-7

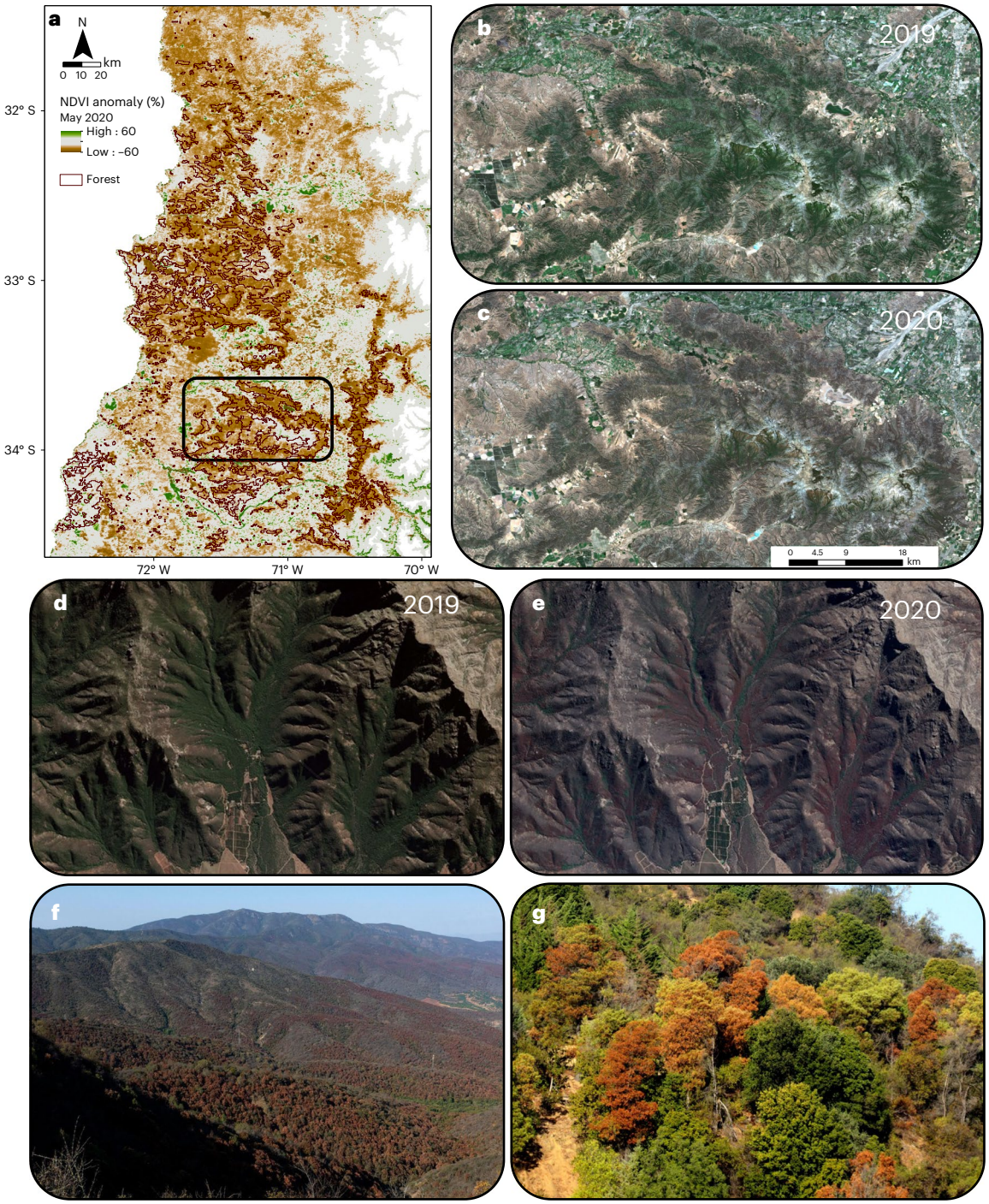


Fig. 3 | Examples showing the magnitude of decline in evergreen sclerophyllous forests in Central Chile. a, NDVI anomaly for May 2020 compared with the mean value for the same month for 2000–2010. **b**,**c**, Sentinel-2 true colour representation of the southern part of Central Chile (boxed area in **a**, 45 km × 70 km) showing the browning effect between 2019 (**b**) and 2020 (**c**). **d**,**e**, Google Earth image from March 2019 (**d**) and March 2020 (**e**) from the same area in Central Chile (5 km × 7 km image). **f**,**g**, Examples of the sclerophyllous forest of Central Chile viewed at landscape (**e**) and plot (**f**) scales. Credit: **f**, Antonio Lara; **g**, Marcelo Miranda.

changes in ecosystem functioning[9,21]. Greater efforts are required to understand the mechanisms behind ecological change, its impacts and the safe operating space for socio-ecological system functioning to prepare for a rapidly changing earth system[32,33]. Among these efforts, in situ interdisciplinary field campaigns may offer clearer understanding of vegetation resilience to climatic extremes, for instance, elucidating whether plants have suffered mortality or simply have been damaged and may subsequently recover. The types of climate anomalies and impacts observed in the past decade in Central Chile suggest that climate change effects, combined with natural variability, particularly if repeated, may be a spyglass to the future for other MTEs.

Methods

Study area

MTEs are characterized mainly by their climate attributes viz. mild wet winters and warm dry summers, and by their vegetation structure

comprising plants with similar functional traits and analogous vegetation types[10]. These ecosystems are highly diverse, having different types of woodland, scrub, pasture land and forest[34]. However, the dominant vegetation is the typical sclerophyllous forest and shrubland, an analogous formation among the MTEs that is associated with high plant species diversity[10].

Five MTE regions are recognized globally: South Africa, the Mediterranean Basin, California, Southwest Australia and Central Chile[10]. The boundaries of these regions are not consistently defined and depend on the delineation approach[35]. To delineate these regions, we combined two criteria: (1) areas classified as Csa or Csb by Köppen–Geiger, obtained from ref. 36 as a 1-km-grid global map and (2) arid and semi-arid areas worldwide using the global aridity index (the mean annual precipitation/mean annual reference evapotranspiration), obtained from ref. 37 (Supplementary Fig. 1). The analysis was done for the forest area intersecting both datasets.

Climatic data and analysis

To provide a global-scale depiction of the precipitation anomalies during the past decade (Fig. 1), we used the Global Precipitation Climatology Project monthly precipitation dataset (1979–present) combining observations and satellite precipiation data into 2.5° × 2.5° grids[38]. The anomalies are departures from the long-term mean for 1980–2010. For a more specific analyses of the precipitation and temperature anomalies in the MTEs, we used the Climate Research Unit (CRU) TS dataset (release 4.04) for January 1901–December 2020 on a 0.5° × 0.5° grid over landmasses[39]. The CRU-TS dataset was derived using interpolation of monthly climate anomalies from extensive networks of weather-station observations. Values for all MTEs (used for Figs. 1b and 2) are the area mean of the grid points within each MTE bounding box.

The SPEI was obtained from the SPEI Global Drought Monitor (https://spei.csic.es/) with monthly data from 1955 onwards available at 1° × 1° grids over land domains. SPEI values for each MTE (used for Supplementary Fig. 2) are the area mean of the grid points within each MTE's bounding box.

To provide an estimate of the precipitation anomalies caused by climate change during the past decade, we considered the 2010–2020 precipitation anomalies from 26 general circulation models comprising the Coupled Model Intercomparison Project Phase 5[40] and integrated them under the representative concentration pathway 8.5 (RCP8.5) scenario. Figure 1b includes the multimodel mean anomalies that effectively isolate the signal due to anthropogenic climate forcing.

Forest change analysis

The evaluated forest data were obtained from the MCD12Q1 V6 MODIS Global Land Cover product via the Google Earth Engine platform (GEE)[41], with a spatial resolution of 500 m. We considered four types of forest: (1) evergreen needleleaf forests: dominated by evergreen conifer trees (canopy >2 m) and tree cover >60%, (2) evergreen broadleaf forests: dominated by evergreen broadleaf and palmate trees (canopy >2 m) and tree cover >60%, (3) mixed forests: dominated by neither deciduous nor evergreen (40–60% of each) tree type (canopy >2 m) and a tree cover >60% and (4) woody savannas: tree cover 30–60% (canopy >2 m)[42]. We excluded all burned pixels within the 2000–2021 period using the monthly global burned area product MCD64A1.006 from MODIS[43]. The code used in GEE is available at https://code.earthengine.google.com/b90dc3d159b8be88dc8aa52308c227d7

Statistical analysis of time series

We applied a time series analysis to assess the changes that the forest has experienced during the past two decades. We used the NDVI data from the MOD13Q1 V6 product (also available in GEE). This global product allows the evaluation of NDVI every 16 days at a spatial resolution of 250 m. NDVI is frequently used as an indication of greenness or productivity of vegetation canopy. While a robust measure of productivity and valid way of studying vegetation response to climate change[14,15], it is nevertheless unable to distinguish directly among processes that drive apparent vegetation change (for example, phenological changes in leaves are not distinguishable from die-off). To avoid the potential for any confounding factors to mediate the signal of NDVI, we restricted our analysis to forested and unburned areas only. First, we masked the pixels of the forest using the MCD12Q1 V6 MODIS 2000 Global Land Cover product for the regional analysis in all MTEs. We extracted the NDVI time series from MOD13Q1 V6 product masked by (1), (2), (3) and (4) detailed above in the 'Forest change analysis' section and excluding the burned areas with the MCD64A1.006 product of MODIS. For the more detailed analysis of Central Chile, we used as an initial mask the land cover of Chile elaborated in ref. 44 for 2014 with 30 m spatial resolution. To exclude forest disturbed areas that could introduce noise in the analysis, we deleted from the initial mask the burned areas from 2000–2019 using the MOD13Q1 V6 product, in addition to the areas affected by tree cover loss using the 30-m-resolution[45] product also available in GEE. Tree cover loss is defined as a stand replacement disturbance that means the removal or mortality of all tree cover in a Landsat pixel with more than 50% of tree crown cover[46]. Finally, to avoid the noise included by adjacent land covers in the time series analysis (for example, annual grasslands, agriculture, shrublands), we selected only the NDVI MODIS pixels (250 m spatial resolution) covered by more than 90% of 30-m-resolution forest pixels. The filtering process left 16,641 NDVI MODIS time series available for Central Chile.

To identify the trend of the time series of NDVI for each MTE, we applied a modified Mann–Kendall test for autocorrelated data in the 'modifiedmk' package, which assesses statistical significance of the temporal changes[45]. The analysis considered the period 21 June 2000 to 21 June 2021 at the original 16 day timescale wherein we assessed the magnitude of the trend via Sen's slope test. The percentage of pixels with a significant negative trend was defined considering a P value of 0.01. All analyses were done in the statistical software R v.4.0.5.

Time series change detection analysis

We conducted a time series change detection analysis using threshold regression models. These models allow for the characterization of different types of threshold effect observed in the time series, providing an interpretable representation of nonlinear relationships between variables. By incorporating change points as threshold parameters, these models enable the identification of non-changing points, abrupt changes and smooth-changing time series[47]. To perform the analysis, we employed the 'chngpt' (Estimation and Hypothesis Testing for Threshold Regression) R software package v.2023.1-30, which can be accessed at https://cran.r-project.org/web/packages/chngpt/index.html. For detailed information on its functionality, please refer to ref. 47. Specifically, we utilized the 'chngptm' function to evaluate change points in the regional mean time series of NDVI for MTEs. Comparison of Akaike information criterion values among the models enabled us to identify the best-fitting model that aligns with the observed data, incorporating the threshold and statistically significant changes[47].

Vegetation field surveys

The vegetation survey was carried out in the Mediterranean region of Chile, which extends from −32.96° to −34.1° S and −70.42° to −71.33° W (Supplementary Fig. 3). This region comprises an east–west gradient composed of three sectors (Andes Range Footshills, Inner Valley and Coastal Range) covered by shrubland and sclerophyll forests under different degrees of degradation.

Field data were used as descriptors of forest composition and productivity at a plot and individual scale to explore the trends towards forest canopy collapse. Individual-focused data were obtained from 474 trees (diameter >10 cm) of mesic species (that is, *C. alba*–*P. boldus*) and xeric

species (that is, *Q. saponaria*–*L. caustica*) dominant in Central Chile Mediterranean forests (Supplementary Table 2). The mesic condition relates to sites where the terrain topography and soil characteristics allow the development of species with high humidity demand. On the other hand, the xeric condition corresponds to sites where the environmental conditions of humidy and temperature restrict the development of species.

Plot data included 110, 78 and 70 circular samples (400 m^2) for the years 2017, 2019 and 2020, respectively, located on the Coastal and Andean Range, and on the Innerl valley (Supplementary Fig. 3).

To assess vegetation productivity, we used NDVI and LAI as proxies for plot and tree function, respectively. These variables reflect the tree's capacity to absorb energy for photosynthesis, maintain leaf turgor and preserve the global energy balance in the canopy. Values for NDVI were calculated for each sample plot as derived from Landsat 8 imagery (resolution 30 m). The NDVI at the plot level was calculated as the mean of the spring season NDVI. Values of LAI at the individual level were obtained from the field realizing four hemispherical measurements under the tree crown (north, south, east and west) using a ceptometer (model DECAGON ACCUPAR LP-80).

Bias-corrected and accelerated bootstrap intervals were calculated to test the difference between medians for LAI, considering all combinations among the years studied. The results of this analysis are presented in Supplementary Table 3.

Due to the unusual drought condition during the spring–summer 2019 and its effect on tree canopy, we recorded the percentage of the crown affected by browning for each individual. All data were measured for the years 2017, 2019 and 2020 during the spring and summer seasons.

Inclusion and ethics statement

All collaborators involved in this study meet the authorship criteria of Nature Portfolio journals. The roles and responsibilities were established and agreed upon by all collaborators. It is noteworthy that all authors hail from the majority of regions under study: Central Chile, the Mediterranean Basin and California, thereby ensuring a comprehensive representation. The significance of this work lies in its relevance at global but also at the local level. This research did not involve the study of human participants.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the article. Source data are provided with this paper.

Code availability

The codes generated during the current study are available from the corresponding author on reasonable request.

References

1. Godfree, R. C. et al. Historical reconstruction unveils the risk of mass mortality and ecosystem collapse during pancontinental megadrought. *Proc. Natl Acad. Sci. USA* **116**, 15580–15589 (2019).
2. Allen, C. D., Breshears, D. D. & McDowell, N. G. On underestimation of global vulnerability to tree mortality and forest die-off from hotter drought in the Anthropocene. *Ecosphere* **6**, 129 (2015).
3. Allen, C. D. et al. A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *For. Ecol. Manage.* **259**, 660–684 (2010).
4. Bozkurt, D., Rojas, M., Boisier, J. P. & Valdivieso, J. Projected hydroclimate changes over Andean basins in central Chile from downscaled CMIP5 models under the low and high emission scenarios. *Climatic Change* **150**, 131–147 (2018).
5. Bergstrom, D. M. et al. Combating ecosystem collapse from the tropics to the Antarctic. *Glob. Change Biol.* **27**, 1692–1703 (2021).
6. Bland, L. M. et al. Developing a standardized definition of ecosystem collapse for risk assessment. *Front. Ecol. Environ.* **16**, 29–36 (2018).
7. Newton, A. C. et al. Operationalising the concept of ecosystem collapse for conservation practice. *Biol. Conserv.* **264**, 109366 (2021).
8. Newton, A. C. *Ecosystem Collapse and Recovery* (Cambridge Univ. Press, 2021).
9. Berdugo, M. et al. Global ecosystem thresholds driven by aridity. *Science* **367**, 787–790 (2020).
10. Rundel, P. W. et al. Mediterranean biomes: evolution of their vegetation, floras, and climate. *Annu. Rev. Ecol. Evol. Syst.* **47**, 383–407 (2016).
11. Anderegg, W. R. L., Anderegg, L. D. L., Kerr, K. L. & Trugman, A. T. Widespread drought-induced tree mortality at dry range edges indicates that climate stress exceeds species' compensating mechanisms. *Glob. Change Biol.* **25**, 3793–3802 (2019).
12. Adams, H. D. et al. A multi-species synthesis of physiological mechanisms in drought-induced tree mortality. *Nat. Ecol. Evol.* **1**, 1285–1291 (2017).
13. Choat, B. et al. Triggers of tree mortality under drought. *Nature* **558**, 531–539 (2018).
14. Das, A. J. et al. Empirically validated drought-vulnerability mapping in the mixed conifer forests of the Sierra Nevada. *Ecol. Appl.* **32**, e2514 (2022).
15. Pettorelli, N. et al. Using the satellite-derived NDVI to assess ecological responses to environmental change. *Trends Ecol. Evol.* **20**, 503–510 (2005).
16. Garreaud, R. D. et al. The Central Chile mega drought (2010–2018): a climate dynamics perspective. *Int. J. Climatol.* **40**, 421–439 (2019).
17. Chen, D. et al. in *Climate Change 2021: The Physical Science Basis* (eds Masson-Delmotte, V. et al) 147–286 (Cambridge Univ. Press, 2021).
18. Garreaud, R. D., Clem, K. & Veloso, J. V. The South Pacific pressure trend dipole and the southern blob. *J. Clim.* **34**, 7661–7676 (2021).
19. Arroyo, M. T. K. et al. Extreme drought affects visitation and seed set in a plant species in the Central Chilean Andes heavily dependent on hummingbird pollination. *Plants* **9**, 1553 (2020).
20. Fettig, C. J., Mortenson, L. A., Bulaon, B. M. & Foulk, P. B. Tree mortality following drought in the central and southern Sierra Nevada, California, U.S. *For. Ecol. Manage.* **432**, 164–178 (2019).
21. Senf, C., Buras, A., Zang, C. S., Rammig, A. & Seidl, R. Excess forest mortality is consistently linked to drought across Europe. *Nat. Commun.* **11**, 6200 (2020).
22. Matusick, G., Ruthrof, K. X., Brouwers, N. C., Dell, B. & Hardy, G. S. J. Sudden forest canopy collapse corresponding with extreme drought and heat in a Mediterranean-type eucalypt forest in southwestern Australia. *Eur. J. Res.* **132**, 497–510 (2013).
23. Andivia, E. et al. Climate and species stress resistance modulate the higher survival of large seedlings in forest restorations worldwide. *Ecol. Appl.* **31**, e02394 (2021).
24. Schenk, H. J. et al. Hydraulic integration and shrub-growth form linked across continental aridity gradients. *Proc. Natl Acad. Sci. USA* **105**, 11248–11253 (2008).
25. Nolan, R. H. et al. Limits to post-fire vegetation recovery under climate change. *Plant Cell Environ.* **44**, 3471–3489 (2021).
26. Holmgren, M. Exotic herbivores as drivers of plant invasion and switch to ecosystem alternative states. *Biol. Invasions* **4**, 25–33 (2002).

27. Chávez, R. O. et al. A probabilistic multi-source remote sensing approach to evaluate extreme precursory drought conditions of a wildfire event in Central Chile. *Front. Environ. Sci.* **10**, 865406 (2022).
28. Smith-Ramírez, C. et al. Recovery of Chilean Mediterranean vegetation after different frequencies of fires. *For. Ecol. Manage.* **485**, 118922 (2021).
29. Venegas-González, A. et al. Sclerophyllous forest tree growth under the influence of a historic mega-drought in the Mediterranean Ecoregion of Chile. *Ecosystems* **26**, 344–361 (2023).
30. Seidl, R. et al. Forest disturbances under climate change. *Nat. Clim. Change* **7**, 395–402 (2017).
31. Saatchi, S. et al. Persistent effects of a severe drought on Amazonian forest canopy. *Proc. Natl Acad. Sci. USA* **110**, 565–570 (2013).
32. Dudney, J. & Suding, K. N. The elusive search for tipping points. *Nat. Ecol. Evol.* **4**, 1449–1450 (2020).
33. Hillebrand, H. et al. Thresholds for ecological responses to global change do not emerge from empirical data. *Nat. Ecol. Evol.* **4**, 1502–1509 (2020).
34. Dinerstein, E. et al. An ecoregion-based approach to protecting half the terrestrial realm. *BioScience* **67**, 534–545 (2017).
35. Ackerly, D. D., Stock, W. D. & Slingsby, J. A. in *Fynbos* (eds Allsopp, N. et al.) 361–376 (Oxford Univ. Press, 2014).
36. Beck, H. E. et al. Present and future Köppen–Geiger climate classification maps at 1 km resolution. *Sci. Data* **5**, 180214 (2018).
37. Trabucco, A. & Zomer, R. Global aridity index and potential evapotranspiration (ET0) climate database v.2. *figshare* https://doi.org/10.6084/m9.figshare.7504448.v3 (2019).
38. Adler, R. F. et al. The Version-2 Global Precipitation Climatology Project (GPCP) monthly precipitation analysis (1979–Present). *J. Hydrometeorol.* **4**, 1147–1167 (2003).
39. Harris, I., Osborn, T. J., Jones, P. & Lister, D. Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Sci. Data* **7**, 109 (2020).
40. Taylor, K. E., Stouffer, R. J. & Meehl, G. A. An overview of CMIP5 and the experiment design. *Bull. Am. Meteorol. Soc.* **93**, 485–498 (2012).
41. Gorelick, N. et al. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* **202**, 18–27 (2017).
42. Friedl, M. & Sulla-Menashe, D. MCD12Q1 MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 500m SIN Grid V006. *EarthData* https://doi.org/10.5067/MODIS/MCD12Q1.006 (2019).
43. Giglio, L., Justice, C., Boschetti, L. & Roy, D. MCD64A1 MODIS/Terra+Aqua Burned Area Monthly L3 Global 500m SIN Grid V006. *EarthData* https://doi.org/10.5067/MODIS/MCD64A1.006 (2015).
44. Zhao, Y. et al. Detailed dynamic land cover mapping of Chile: accuracy improvement by integrating multi-temporal data. *Remote Sens. Environ.* **183**, 170–185 (2016).
45. Miranda, A. et al. Forest browning trends in response to drought in a highly threatened Mediterranean landscape of South America. *Ecol. Indic.* **115**, 106401 (2020).
46. Hansen, M. C. et al. High-resolution global maps of 21st-century forest cover change. *Science* **342**, 850–853 (2013).
47. Fong, Y., Huang, Y., Gilbert, P. B. & Permar, S. R. chngpt: threshold regression model estimation and inference. *BMC Bioinformatics* **18**, 454 (2017).

Acknowledgements

A.M., S.G, A.L. and R.G. thank ANID/FONDAP/15110009, and A.M. thanks ANID Postdoctoral Fondecyt project 3210101. F.S. thanks ANID grants FB210006 and ACE210006 to the Institute of Ecology and Biodiversity (IEB). A.M. and J.C. acknowledges the support of the European Union's Horizon 2020 research and innovation programme under grant agreement no. 101037419, and J.C. acknowledges Postdoctoral Fondecyt project 3210311. S.G. thanks FORPES project PID2019-106908RA-I00/AEI/10.13039/501100011033 from MICINN, Spain. J.F.O. was supported by the Chilean Foundation of Science and Technology (FONDECYT) Grant 11191147. We also thank the Center of Applied Ecology and Sustainability (CAPES) project PIA/BASAL FB0002. S.V. thanks ANID no. 9219/2022 Concurso Subvención a la Instalación en la Academia, Code 85220080.

Author contributions

A.M., R.G., A.D.S., M.B., A.L. and S.G.-G. conceptualized the project. A.M., R.G. and J.C. developed the methodology. A.M., R.G., R.M., J.C. and M.D.M. curated data. A.M., R.G., R.M., J.C. and M.D.M. conducted formal analysis. A.M., R.G., R.M. and M.D.M. performed visualization. A.M., R.G., F.A.S., M.D.M., J.F.O., C.A.D. and A.L. acquired funding. A.M., J.F.O., C.A.D. and S.V. administered the project. A.M., R.G., A.D.S., M.B., M.D.M., S.G. and A.L. wrote the original draft, which was reviewed and edited by A.M., R.G., J.F.O., A.D.S., M.B., F.A.S., C.A.D., S.V., M.D.M., C.D., S.G. and A.L.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41477-023-01541-7.

Correspondence and requests for materials should be addressed to Alejandro Miranda.

Peer review information *Nature Plants* thanks Víctor Resco de Dios and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2023

nature portfolio

Corresponding author(s): Alejandro Miranda

Last updated by author(s): Aug 6, 2022

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	We use Google Earth Engine to collect data of (1) and (2), and direct download of (3): (1) ndvi temporal series, (2) regional ndvi and burned forest, (3) precipitation, temperature and SPEI for all Mediterranean Type Ecosystems
Data analysis	We use the statistical software R version 4.0.5 package modifiedmk and chngpt for (1), (2) and (3) and Matlab R2021b for (4): (1) ndvi temporal series, (2) regional ndvi, (3) field plot and (4) precipitation, temperature and SPEI for all Mediterranean Type Ecosystems

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

(1) ndvi temporal series, (2) regional ndvi, (3) precipitation, temperature and SPEI for all Mediterranean Type Ecosystems

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	We assessed climatic and productivity trends across the world's five Mediterranean forest ecosystems from 2000 – 2021
Research sample	We considered four types of forests for the five Mediterranean Type Ecosystem on the globe: (i) Evergreen Needleleaf Forests: dominated by evergreen conifer trees (canopy >2m) and tree cover >60%, (ii) Evergreen Broadleaf Forests: dominated by evergreen broadleaf and palmate trees (canopy >2m) and tree cover >60%, (iii) Mixed Forests: dominated by neither deciduous nor evergreen (40-60% of each) tree type (canopy >2m) and a tree cover >60%, and (iv) Woody Savannas: tree cover 30-60% (canopy >2m)
Sampling strategy	We use all time series available excluding the burned pixels
Data collection	Data collection was carried out with the Google Earth Engine cloud computing platform.
Timing and spatial scale	We use data from 2000 to 2021
Data exclusions	We exclude burned pixels between 2000 and 2021
Reproducibility	All data used are public and available to everyone.
Randomization	n/a because we use all data, grouping only by the type of Mediterranean type ecosystem
Blinding	n/a

Did the study involve field work? ☒ Yes ☐ No

Field work, collection and transport

Field conditions	The study was conducted in the Central Mediterranean region of Chile, which extends from -32.92° to -34.29°S and -69.76° to -71.71° W. This region is composed by shrubland and sclerophyll forests with different stand structures, located in an east-west gradient comprised by three sectors: Andes Range Footshills, Santiago Basin and Coastal Range (4,874 km²). The climate of the region is temperate Mediterranean (Csb2) with marked dry and warm summer seasons (Peel et al. 2007), with mean temperatures in warmer and colder month of 29.4°C and 14.3°C respectively and a mean annual rainfall of 300 mm year-1. Soils are mainly Alfisols that have an ochric epipedon, an argillic horizon, and moderate to high base saturation. These soils are located in hillocks closer to the Andes Mountains and Coastal Range with slopes from 2–5 % to 20–30 % and are moderately deep to deep and medium to fine-textured.
Location	Central Chile
Access & import/export	Before to the fieldwork (forest inventory), owners and authorities were contacted to obtain authorization for access to the sample

Access & import/export	sites. In the field, the access to each sample plot was realized using paths and tracks available in the field. No vegetal, animal, or soil materials were collected or transported outside the fields.
Disturbance	The forest was not disturbed

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging